Simulations for Everyone! 🎉

mostly.ai/blog/simulations-for-everyone-🎉

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TL;DR: Simulations are powerful. Simulations are hard. They are not for Everyone. Until now!

MOSTLY AI's open-source **Synthetic Data SDK** makes it easy to train Generative AI models on proprietary data. Tens of thousands of users already rely on it for privacy-preserving data sharing across teams, borders, and organizations.

But the full potential of synthetic data goes far beyond privacy. Synthetic data doesn't just show you **how things are** - it can also reveal **how things could be**.

With just a few prompts on the MOSTLY AI platform, users can now run realistic **simulations for any scenario** they choose. These simulations create rich synthetic data that's ready to analyze and easy to understand. They not only predict what might happen, but also show the full range of possible outcomes - along with the probabilities behind them - giving you the insights needed for confident, data-driven decisions.

A Decathlon of Possibilities#

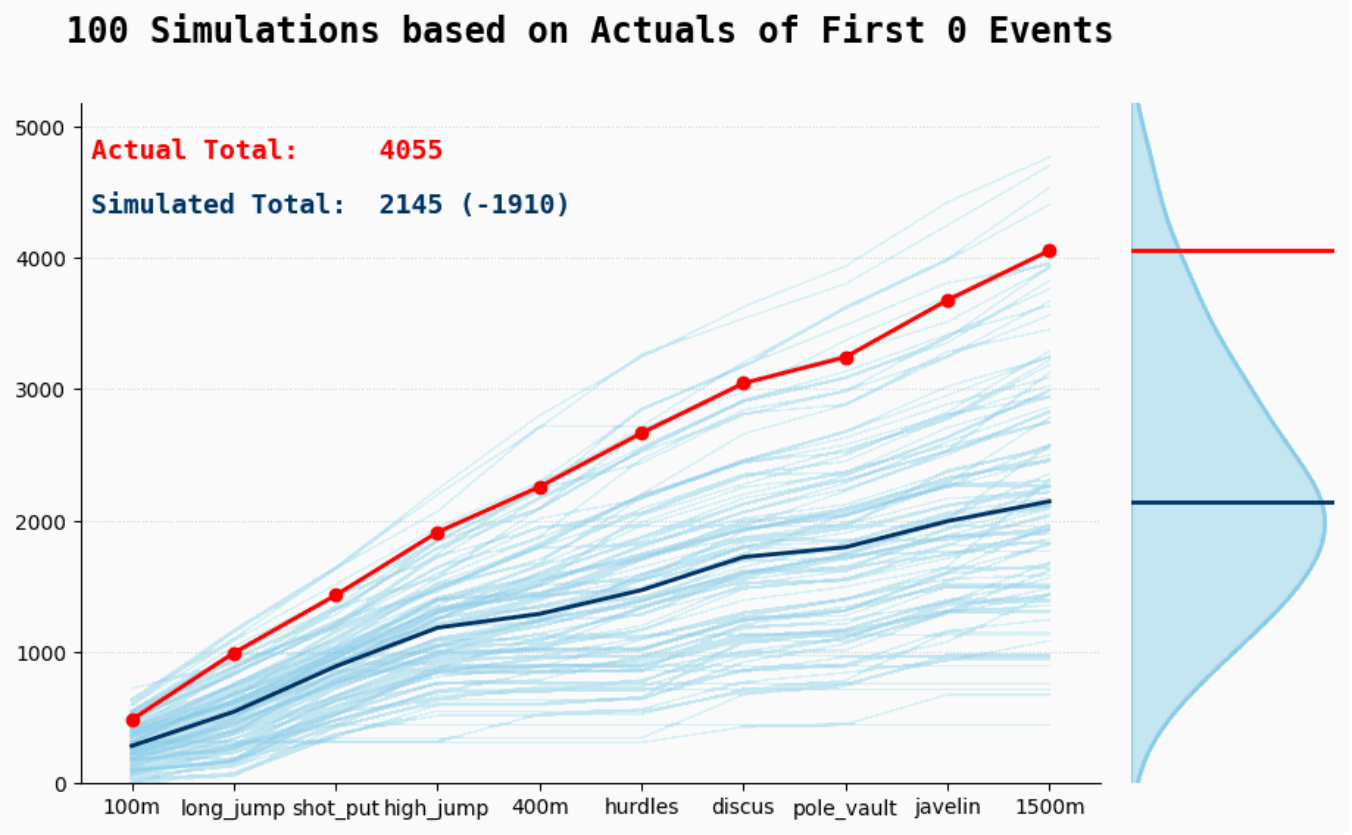
Let's explore a simple example: the decathlon, a track & field event composed of 10 competitions - 100m sprint, javelin, pole vault, and more. Each event contributes to an athlete's total score.

Athletes naturally vary in strengths. Some dominate speed; others shine in strength. So here's the question:

- Can we simulate the **final score** based on results from the first few events?
- Can we see not just a prediction, but the **range of possible outcomes**?
- Can we **run what-if scenarios** by modifying early results?

✅ The answer is: **Yes, Yes and Yes** - and **it's never been easier**!

As training data, we've gathered 30 years of historic decathlon results for hobbyists (downloadable from here as CSV). This was then used for training a generative model that can be explored by anyone and everyone via the MOSTLY AI Assistant. **For example, below you can see 100 simulations each, based on the known outcomes of the first *X* events for a specific athlete.** What's striking is how effectively the generative model narrows its estimates as more information becomes available. While the initial projection spans a wide range—from roughly 500 to 5000 points—it quickly converges. After just three events, we already gain a solid understanding of the likely final score, with uncertainty decreasing significantly as the event series progresses.

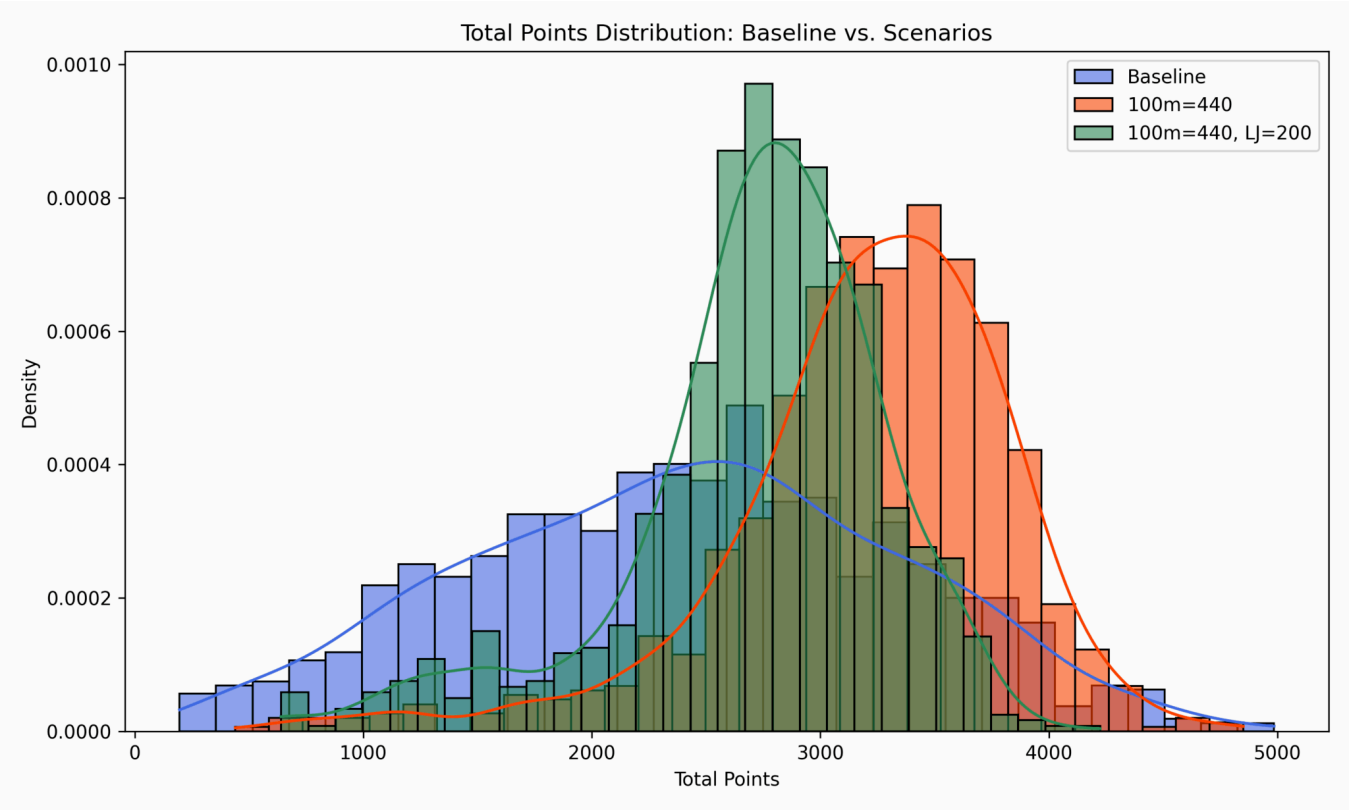


Finally Everyone Can Ask Anything#

Simulations have long been seen as powerful but intimidating - requiring complex math, custom models, and expert knowledge. But that's no longer the case. With MOSTLY AI's Data Intelligence Platform, anyone can run these kinds of advanced simulations in a matter of seconds - **no code, no hassle**. It's simulation made simple. See for yourself:

In the video above, we walk you through the end-to-end process:

1. Train a synthetic data generator using historical records
2. Define a scenario - e.g., early event scores in a decathlon
3. Simulate thousands of synthetic outcomes
4. Analyze and visualize the full range of possibilities



Want to try it yourself? Head to the MOSTLY AI platform to launch your own simulations on top of an already trained generator. The AI Assistant will guide you through the process and help visualize insights at ease.

Beyond Sports: Simulate Your World#

But the power of data-based simulation applies of course far beyond sports. Any domain with rich individual-level data is ripe for providing instant insights via synthetic simulations:

- **Customer Analysis**: Predict purchasing paths based on early interactions
- **Healthcare**: Simulate patient recovery under different treatments
- **Maintenance**: Anticipate failure risks of machines given sensor readings
- **Education**: Forecast scholar performance based on partial test results
- **Planning**: Model network traffic and stress-test conditions
- and many more

As long as historical, individual-level data is available, you can train a generative model to explore “what-if” scenarios and future paths.

Know the Limits: Data-Based, Not Rule-Based#

These data-based simulations are incredibly powerful - but also important to understand:

- They are **data-driven**, not rule-based. These simulations reflect patterns observed in real-world data - not physics engines or predefined systems.
- They are best suited for scenarios **within the realm of observed behavior**. They won't extrapolate to extreme outliers, completely novel scenarios, or complex interactions involving multiple agents or networks.

Think of this as a new lens on reality - one grounded in your data, with full visibility into possible futures.

Simulate is the Future!#

We believe this new capability - available through the Synthetic Data SDK and the MOSTLY AI platform - opens up transformative opportunities for analysts, researchers, and decision-makers across every industry.

Simulations are no longer the domain of specialists. With MOSTLY AI, they belong to everyone.

Try it yourself. Simulate your data. Discover your future.